50 SQL Interview Queries

1. Find duplicate records in a table

```sql
SELECT column1, column2, COUNT(*)
FROM your_table
GROUP BY column1, column2
HAVING COUNT(*) > 1;
```

amazon

2. Retrieve the second highest salary from the Employee table

```sql
SELECT MAX(salary) AS
SecondHighestSalary
FROM Employee
WHERE salary < (SELECT MAX(salary)
FROM Employee);
```

Microsoft

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3. Find employees without department (Left Join usage)

Uber

```sql
SELECT e.*
FROM Employee e
LEFT JOIN Department d
ON e.department_id =
d.department_id
WHERE d.department_id IS NULL;
```

4. Calculate the total revenue per product

```sql
SELECT product_id,
SUM(quantity * price) AS
total_revenue
FROM Sales
GROUP BY product_id;
```

PayPal

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5. Get the top 3 highest-paid employees.

```sql
SELECT TOP 3 *
FROM Employee
ORDER BY salary DESC;
```

Google

6. Customers who made purchases but never returned products.

```sql
SELECT DISTINCT c.customer_id
FROM Customers c
JOIN Orders o ON c.customer_id =
o.customer_id
WHERE c.customer_id NOT IN (
SELECT customer_id FROM Returns
);
```

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7. Show the count of orders per customer.

```sql
SELECT customer_id,
COUNT(*) AS order_count
FROM Orders
GROUP BY customer_id;
```

Meta

8. Retrieve all employees who joined in 2023.

```sql
SELECT *
FROM Employee
WHERE YEAR(hire_date) = 2023;
```

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9. Calculate the average order value per customer.

```
SELECT    customer_id,
AVG(total_amount)   AS
avg_order_value   FROM
Orders     GROUP     BY
customer_id;
```

Microsoft

10. Get the latest order placed by each customer.

```
SELECT customer_id,
MAX(order_date) AS
latest_order_date
FROM Orders
GROUP BY customer_id;
```

Uber

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11. Find products that were never sold.

```sql
SELECT p.product_id
FROM Products p
LEFT JOIN Sales s
ON p.product_id = s.product_id
WHERE s.product_id IS NULL;
```

Adobe

12. Identify the most selling product.

```sql
SELECT TOP 1 product_id,
SUM(quantity) AS total_qty
FROM Sales
GROUP BY product_id
ORDER BY total_qty DESC;
```

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13. Get the total revenue and the number of orders per region.

```sql
SELECT region,
SUM(total_amount) AS total_revenue,
COUNT(*) AS order_count
FROM Orders
GROUP BY region;
```

Meta

14. Count how many customers placed more than 5 orders.

```sql
SELECT COUNT(*) AS customer_count
FROM (
    SELECT customer_id FROM Orders
    GROUP BY customer_id
    HAVING COUNT(*) > 5
) AS subquery;
```

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15. Retrieve customers with orders above the average order value.

```sql
SELECT *
FROM Orders
WHERE total_amount >
(SELECT AVG(total_amount)
FROM Orders);
```

PayPal

16. Find all employees hired on weekends.

```sql
SELECT *
FROM Employee
WHERE DATENAME(WEEKDAY, hire_date)
IN ('Saturday', 'Sunday');
```

Google

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17. Find all employees hired on weekends.

```sql
SELECT *
FROM Employee
WHERE salary BETWEEN 50000 AND
100000;
```

Microsoft

18. Get monthly sales revenue and order count.

```sql
SELECT
FORMAT(date, 'yyyy-MM') AS month,
SUM(amount) AS total_revenue,
COUNT(order_id) AS order_count
FROM Orders
GROUP BY
FORMAT(date, 'yyyy-MM');
```

Google

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19. Rank employees by salary within each department.

```sql
SELECT employee_id, department_id,
salary, RANK() OVER (PARTITION BY
department_id
ORDER BY salary DESC) AS salary_rk
FROM Employee;
```

amazon

20. Find customers who placed orders every month in 2023.

Meta

```sql
SELECT customer_id
FROM Orders
WHERE YEAR(order_date) = 2023
GROUP BY customer_id
HAVING COUNT(DISTINCT
FORMAT(order_date,'yyyy-MM')) = 12
```

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21. Find moving average of sales over the last 3 days.

```sql
SELECT order_date,
AVG(total_amount) OVER (ORDER BY
order_date ROWS BETWEEN 2 PRECEDING
AND CURRENT ROW) AS moving_avg
FROM Orders;
```

Microsoft

22. Identify the first and last order date for each customer.

```sql
SELECT customer_id,
MIN(order_date) AS first_order,
MAX(order_date) AS last_order
FROM Orders
GROUP BY customer_id;
```

Uber

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23. Show product sales distribution (percent of total revenue).

```sql
WITH TotalRevenue AS (
SELECT
SUM(quantity * price) AS total FROM Sales)
SELECT s.product_id,
SUM(s.quantity * s.price) AS revenue,
SUM(s.quantity * s.price) * 100/ t.total
AS revenue_pct
FROM Sales s
CROSS JOIN TotalRevenue t
GROUP BY s.product_id, t.total;
```

PayPal

24. Retrieve customers who made consecutive purchases (2 Days)

Walmart

```sql
WITH cte AS (
SELECT id, order_date,
LAG(order_date) OVER (PARTITION BY id
ORDER BY order_date) AS prev_order_date
FROM Orders)
SELECT id, order_date, prev_odate
FROM cte
WHERE
DATEDIFF(DAY, prev_odate, order_date) = 1;
```

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24. Retrieve customers who made consecutive purchases (2 Days)

Walmart

```
WITH cte AS (
SELECT id, order_date,
LAG(order_date) OVER (PARTITION BY id
ORDER BY order_date) AS prev_order_date
FROM Orders)
SELECT id, order_date, prev_odate
FROM cte
WHERE
DATEDIFF(DAY, prev_odate, order_date) = 1;
```

25. Find churned customers (no orders in the last 6 months).

```
SELECT customer_id
FROM Orders
GROUP BY customer_id
HAVING
MAX(order_date) <
DATEADD(MONTH,-6,GETDATE());
```

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26. Calculate cumulative revenue by day.

```sql
SELECT order_date,
SUM(total_amount) OVER
(ORDER BY order_date) AS
cumulative_revenue
FROM Orders;
```

Adobe

27. Identify top-performing departments by average salary.

```sql
SELECT department_id,
AVG(salary) AS avg_salary
FROM Employee
GROUP BY department_id
ORDER BY avg_salary DESC;
```

Google

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28. Find customers who ordered more than the average number of orders per customer.

Meta

```
WITH customer_orders AS (
SELECT customer_id, COUNT(*) AS order_count
FROM Orders
GROUP BY customer_id)
SELECT * FROM customer_orders
WHERE order_count > (SELECT
AVG(order_count) FROM customer_orders);
```

29. Calculate revenue generated from new customers (first-time orders).

Microsoft

```
WITH first_orders AS (
SELECT customer_id, MIN(order_date) AS
first_order_date FROM Orders
GROUP BY customer_id)
SELECT SUM(o.total_amount) AS new_revenue
FROM Orders o JOIN first_orders f
ON o.customer_id = f.customer_id
WHERE o.order_date = f.first_order_date;
```

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30. Find the percentage of employees in each department.

```sql
SELECT
  department_id,
 COUNT(*) AS emp_count,
COUNT(*) * 100.0 / (SELECT
COUNT(*) FROM Employee)
AS pct FROM Employee
GROUP BY department_id;
```

Uber

31. Retrieve the maximum salary difference within each department.

```sql
SELECT
department_id,
MAX(salary) - MIN(salary) AS
salary_diff
FROM Employee
GROUP BY department_id;
```

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32. Find products that contribute to 80% of the revenue (Pareto Principle).

```sql
WITH sales_cte AS (
SELECT product_id, SUM(qty * price) AS revenue
FROM Sales GROUP BY product_id),
total_revenue AS (
SELECT SUM(revenue) AS total FROM sales_cte)
SELECT s.product_id, s.revenue,
SUM(s.revenue) OVER
(ORDER BY s.revenue DESC ROWS BETWEEN UNBOUNDED
PRECEDING AND CURRENT ROW) AS running_total
FROM sales_cte s, total_revenue t
WHERE SUM(s.revenue) OVER (ORDER BY s.revenue DESC
ROWS BETWEEN UNBOUNDED PRECEDING AND
CURRENT ROW) <= t.total * 0.8;
```

Walmart

33. Calculate average time between two purchases for each customer.

```sql
WITH cte AS (
SELECT customer_id, order_date,
LAG(order_date) OVER (PARTITION BY
customer_id
ORDER BY order_date) AS prev_date
FROM Orders)
SELECT customer_id,
AVG(DATEDIFF(DAY, prev_date, order_date))
AS avg_gap_days FROM cte
WHERE prev_date IS NOT NULL
GROUP BY customer_id;
```

Meta

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34. Show last purchase for each customer along with order amount.

```sql
WITH ranked_orders AS
(SELECT customer_id, order_id,
total_amount, ROW_NUMBER() OVER
(PARTITION BY customer_id ORDER BY
order_date DESC) AS rn FROM Orders)
SELECT customer_id, order_id,
total_amount
FROM ranked_orders
WHERE rn = 1;
```

Google

35. Calculate year-over-year growth in revenue.

Microsoft

```sql
SELECT FORMAT(order_date, 'yyyy') AS year,
SUM(total_amount) AS revenue,
SUM(total_amount) - LAG(SUM(total_amount))
OVER (ORDER BY FORMAT(order_date, 'yyyy'))
AS yoy_growth
FROM Orders
GROUP BY FORMAT(order_date, 'yyyy');
```

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36. Detect customers whose purchase amount is higher than their historical 90th percentile.

```sql
WITH ranked_orders AS (
 SELECT customer_id, order_id,
total_amount,
NTILE(10) OVER (PARTITION BY customer_id
ORDER BY total_amount) AS decile
FROM Orders)
SELECT customer_id, order_id, total_amount
FROM ranked_orders
WHERE decile = 10;
```

amazon

37. Retrieve the longest gap between orders for each customer.

```sql
WITH cte AS (
SELECT customer_id, order_date,
LAG(order_date) OVER (PARTITION BY
customer_id ORDER BY order_date) AS
prev_order_date FROM Orders)
SELECT customer_id, MAX(DATEDIFF(DAY,
prev_order_date, order_date)) AS max_gap
FROM cte
WHERE prev_order_date IS NOT NULL
GROUP BY customer_id;
```

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38. Identify customers with revenue below the 10th percentile.

Google

```sql
WITH cte AS (
SELECT customer_id, SUM(total_amount) AS
total_revenue
FROM Orders
GROUP BY customer_id)
SELECT customer_id, total_revenue
FROM cte
WHERE total_revenue <
(SELECT PERCENTILE_CONT(0.1) WITHIN GROUP
(ORDER BY total_revenue) FROM cte);
```

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